

DFA $M = (Q, \Sigma, s, \delta, F)$

$\delta: Q \times \Sigma \rightarrow Q$. Transition function given in the definition of M .

Extending δ to strings.

$$\hat{\delta}: Q \times \Sigma^* \rightarrow Q.$$

$$\hat{\delta}(q, \epsilon) = q$$

$$\hat{\delta}(q, xa) = \delta(\underbrace{\hat{\delta}(q, x)}_{\text{I.H.}}, a).$$

x is accepted by M if $\hat{\delta}(s, x) \in F$

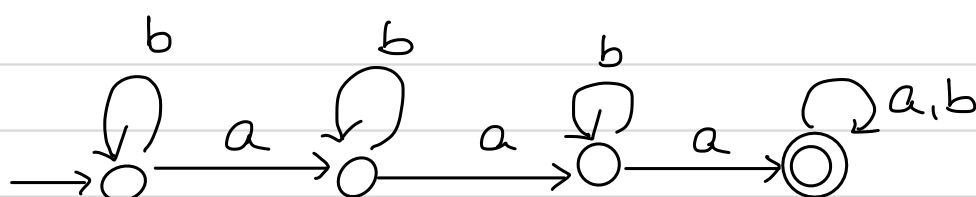
x is rejected by M if $\hat{\delta}(s, x) \notin F$

Language accepted by M

$$L(M) = \{x \in \Sigma^* \mid \hat{\delta}(s, x) \in F\}.$$

$A \subseteq \Sigma^*$ is regular if $A = L(M)$ for some DFA M .

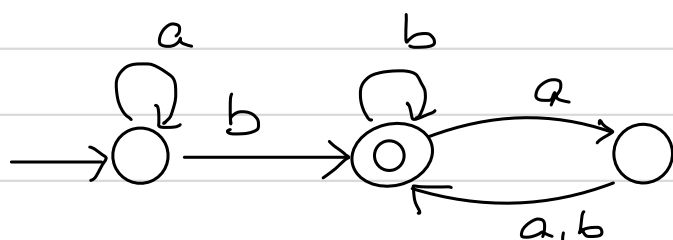
Example



$x: b a a b b a a b \in L(M)$

$y: b a b b b a b \notin L(M)$.

$L(M) = \{x \in \Sigma^* \mid x \text{ contains 3 or more } a\text{'s}\}$.



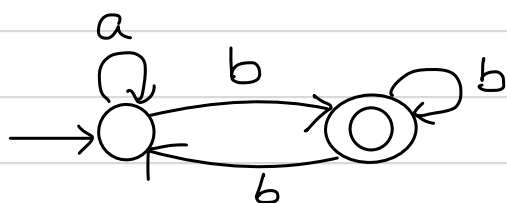
$a a b b b \in L(M)$

$a a b \in L(M)$

$a b a a \in L(M)$

$a b a a a \notin L(M)$

$L(M) = \{x \in \Sigma^* \mid x \text{ contains at least one } b \text{ and an even number of } a\text{'s follow the last } b\}$.



$L(M) = \{x \in \Sigma^* \mid x \text{ ends in } b\}$



Example with proof of construction.

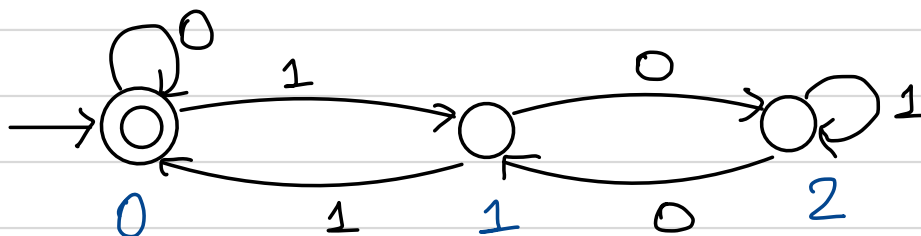
$A = \{ x \in \{0,1\}^* \mid x \text{ represents a multiple of three in binary} \}$.

- leading zeros permitted.
- ϵ - represents 0.

A is regular. Construct a DFA M s.t. $L(M) = A$.

in A	0	0
	11	3
	110	6
	1001	9
	1100	12

M :



Theorem. $L(M) = A$.

Let $\#x$ - the number represented by string x in binary.

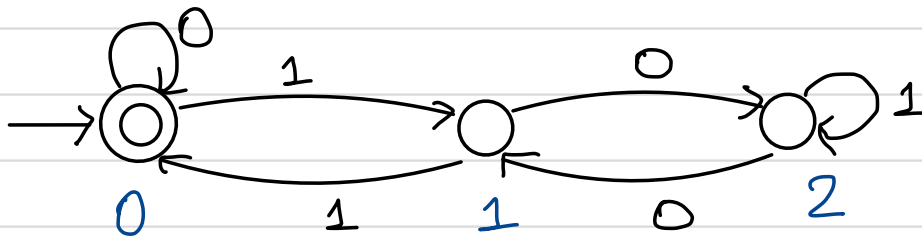
$$\# \epsilon = 0$$

$$\# 0 = 0$$

$$\# 11 = 3$$

$A = \{x \in \{0,1\}^* \mid x \text{ represents a multiple of three in binary}\}$.

M:



For $x \in \{0,1\}^*$

$$\begin{aligned} \hat{S}(0, x) &= 0 \text{ iff } \#x \equiv 0 \pmod{3} \\ \hat{S}(0, x) &= 1 \text{ iff } \#x \equiv 1 \pmod{3} \\ \hat{S}(0, x) &= 2 \text{ iff } \#x \equiv 2 \pmod{3} \end{aligned}$$

To prove: $\hat{S}(0, x) = \#x \pmod{3}$.

$$\left. \begin{aligned} \#(x0) &= 2(\#x) + 0 \\ \#(x1) &= 2(\#x) + 1 \end{aligned} \right\} \begin{aligned} \#(xc) &= 2(\#x) + c \\ &\forall c \in \{0,1\}. \end{aligned} \quad \textcircled{B}$$

For all $q \in \{0,1,2\}$ and symbol $c \in \{0,1\}$.

$$S(q, c) = (2q + c) \pmod{3}. \quad \text{---} \quad \textcircled{A}$$

Induction on $|x|$.

$$\begin{aligned} \hat{S}(0, xcc) &= S(\hat{S}(0, x), c) \\ &\quad \text{[Def of } \hat{S}] \\ &= S(\#x \pmod{3}, c) \quad \text{[IH]} \end{aligned} \quad \left| \quad \begin{aligned} \text{Base Case } x = \epsilon. \\ \hat{S}(0, \epsilon) &= 0 \text{ [def. of } \hat{S}] \\ &= \#\epsilon \\ &= \#\epsilon \pmod{3}. \end{aligned} \right.$$

$$= (2(\#x \pmod{3}) + c) \pmod{3} \quad \text{[From } \textcircled{A}].$$

$$= (2(\#x) + c) \pmod{3}.$$

$$= \#x c \pmod{3}. \quad \text{[From } \textcircled{B}]$$

Closure properties of regular sets.

Let $A, B \subseteq \Sigma^*$

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\} \quad \text{Union}$$

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\} \quad \text{Intersection}$$

$$\overline{A} = \{x \in \Sigma^* \mid x \notin A\} \quad \text{Complement}$$

Question. Are regular set closed under
union, intersection, complementation?

Intersection:

if A and B are regular then is $A \cap B$ regular?

Let $A, B \subseteq \Sigma^*$

Statement: if A and B are regular then $A \cap B$ is regular.

$$\exists M_1 \text{ s.t. } L(M_1) = A ; \exists M_2 \text{ s.t. } L(M_2) = B$$

To construct M_3 s.t. $L(M_3) = A \cap B$.

M_3 runs over strings in Σ^* ; so $L(M_3) \subseteq \Sigma^*$

$$M_1 = (Q_1, \Sigma, \delta_1, \delta_1, F_1) \quad M_2 = (Q_2, \Sigma, \delta_2, \delta_2, F_2)$$
$$M_3 = (Q_3, \Sigma, \delta_3, \delta_3, F_3).$$

$$Q_3 = Q_1 \times Q_2 = \{ (q_1, q_2) \mid q_1 \in Q_1 \text{ and } q_2 \in Q_2 \}.$$

$$\delta_3 = (\delta_1, \delta_2)$$

$$F_3 = F_1 \times F_2 = \{ (q_1, q_2) \mid q_1 \in F_1 \text{ and } q_2 \in F_2 \}.$$

$$\delta_3: Q_3 \times \Sigma \rightarrow Q_3$$

$$\delta_3(\underbrace{(q_1, q_2)}_{\in Q_3}, a) = \underbrace{ \left(\overbrace{\delta_1(q_1, a)}^{\in Q_1}, \overbrace{\delta_2(q_2, a)}^{\in Q_2} \right) }_{\in Q_3}$$

M_3 - product of M_1 & M_2 .